R13

Code No: 115DT

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2017
COMPUTER NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a)	Write the advantages of optical fiber over twisted-pair and coaxial cables.	[2]
b)	What are the advantages of having layered architecture?	[3]
c)	Briefly explain the difference between switch and router.	[2]
d)	Sketch the Manchester encoding for the bit stream: 0001110101.	[3]
e)	Give the advantages of hierarchical routing.	[2]
f)	Differences between CO and CL.	[3]
g)	Explain DHCP.	[2]
h)	What are the functions of ICMP?	[3]
i)	What is the architecture of WWW?	[2]
j)	Explain the differences between POP3 and IMAP.	[3]

PART - B

(50 Marks)

2.a)	Compare and contrast the OSI and TCP/IP reference models.	
b)	What are the different types of error detection methods? Explain the CRC error detection technique using generator polynomial x^4+x^3+1 and data 11100011.	[5+5]
	OR	
3.a)	Discuss about the various transmission media available at the physical layer.	
b)	Explain about GBN Sliding Window Protocol.	[5+5]
4.a)	Explain the differences between the switching methods.	
b)	Elucidate the CSMA schemes.	[5+5]
	OR	
5.a)	Illustrate the frame structure of IEEE 802.3.	
b)	Give a detail note on the ALOHA protocols.	[5+5]
6.a)	Elucidate Distance Vector Routing Algorithm with example.	
b)	Describe the problem and solutions associated with distance vector routing.	[5+5]
	OR	
7.a)	Explain the general principles of congestion control.	
b)	Describe congestion control in datagram subnets.	[5+5]

8.a) Elucidate the special IP addresses used in internet.
b) Discuss the significance and the operation of NAT. [5+5]

OR

9.a) Illustrate the connection establishment and release in transport layer.
b) How crash recovery is managed at the transport layer? [5+5]

10.a) Explain Real-time transport protocol.
b) When user clicks a hyperlink, what are the steps that occur between the user's click and the page being displayed? [5+5]

OR

11. Write short notes on the following: [10]
(a) MIME (b) Audio compression (c) DNS (d) Voice over IP.

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